Verheul’s printed distortion map: actual disproof

Result. Lean disproves the y-scaling coordinate formula in Verheul (2004), §4.1, p.288, using an actual nonsingular curve and point within the printed prime bounds. The formula sends P=(4,6) off $y^2=x^3+1$ over an actual extension of F29. The corrected x-scaling formula is proved as an actual point-group automorphism, with explicit inverse and period three. Signed standing unverified. [pcc:unverified]

Actual E/F29: $y^2=x^3+1$, #E=30, P=(4,6) of order5

Printed: (4,6w)	Repaired: (4w,6)	Control: (0,1)
Not a curve point	Order5; outside ⟨P⟩	Order3; fixed by repair

Actual F841, primitive cube root w outside F29; p=29 and ℓ=5 both exceed3.
Corrected point-group map T(x,y)=(wx,y): addition preserved, T^3=identity.

Generic proof. For every field, nontrivial cube root w and curve point with nonzero ordinate, multiplying y by w violates the equation: cancellation would force $w^2=1$, contradicting $w^3=1$ and $w\neq1$. Multiplying x by w preserves the equation. Under nonzero discriminant, Lean derives nonsingularity and proves actual transformed slope/addition identities in all branches, then constructs the point-group homomorphism, bijection and threefold identity. [pcc:unverified]

Concrete qualified instance. F841 is the actual quadratic algebra over F29 with generator satisfying $w^2+w+1=0$; its field criterion, cardinality841, primitive cube root and exclusion from every base scalar are checked. The actual base curve has30points by an explicit point/code equivalence and kernel count, giving source trace zero. Both base and extension points (4,6) have exact order5. Both primes exceed3, $29\equiv2 \bmod 3$, gcd(29,5)=1, and the cyclotomic degree is two. The printed output is not even a curve point, so no point map can have those output coordinates at P. [pcc:unverified]

Distortion and negative control. The actual repaired image Q has exact order5 and lies outside the actual cyclic subgroup generated by P. Every nonzero member of that subgroup is mapped outside it; integer cyclic membership is covered, not just a bounded list of natural scalars. In contrast, actual (0,1) has order3 and is fixed by the repair. Thus the corrected map is not claimed to distort every nonzero point. [pcc:unverified]

Primary attribution. E.R.Verheul, Journal of Cryptology17(2004),277–296, DOI10.1007/s00145-004-0313-x. Publisher PDF444741B/SHA83f55f48…4a14da0 and author PDF244822B/SHA28dffc31…0f42afe agree on the coordinate. Global bounds, relevant statements/proofs and original Theorem5 were read; publisher pp288/289 visually inspected. Additive coverage retains exact read scope; no full-paper read, frozen backlog edit, canonical status promotion or novelty claim. [pcc:unverified]

Validation. Whole gate 49 declarations /0 failures /0 trusted native exceptions; pinned Lean4.35.0/Mathlib56a00585…. Complete closed binding and custom source/config/artifact pins are retained, with lossless failed/successful attempts and full gate. Only propext/Classical.choice/Quot.sound; no holes, custom axioms, native evaluator, raised limits, new global flags or waivers. Exact PCC replay is separate; signed standing unverified. [pcc:unverified]

Remaining scope. The explicit printed coordinate is refuted; the general distortion-existence Theorem5 is not. Endomorphism-ring/Tate-module representation, canonical Weil/Tate pairing values, scheme/isogeny formalization and general geometric supersingularity remain open. The actual count verifies the source trace predicate. The corpus’s wider Tate-value paraphrase receives no automatic promotion. KB unavailable; bounded primary/formal fallback recorded. No native experiment, paid compute or agent upload. Other load-bearing papers and all-fourteen obligations remain active. [pcc:unverified]